

Project Report: Sales Funnel Optimization & Lead Conversion Analysis

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Abstract

This report presents a comprehensive analysis of Knovist Technologies' lead conversion and sales data. The primary objective was to identify structural bottlenecks in the sales funnel, evaluate the effectiveness of marketing channels and sales personnel, and build a predictive model to identify high-value leads. Key findings reveal a critical funnel bottleneck where approximately 24.5% of leads are dropped before a single outbound call attempt, stark variance in sales representative performance, and significant differences in cost-effectiveness across marketing channels. A CatBoost classifier was successfully trained to predict lead conversion with 96.8% accuracy, providing a powerful tool for lead prioritization.

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1 Executive Summary

This report presents a comprehensive analysis of Knovist Technologies' lead conversion and sales data. The primary objective was to identify structural bottlenecks in the sales funnel, evaluate the effectiveness of marketing channels and sales personnel, and build a predictive model to identify high-value leads.

Key Findings:

- **Critical Funnel Bottleneck:** Approximately **24.5% of leads** (6,729 records) are "dropped" before a single outbound call is attempted, representing a significant waste of marketing spend.
- **Sales Performance Variance:** A stark performance gap exists between top-tier sales representatives (e.g., Varsha A V, Sneha Rajesh) with ~50% conversion rates and lower-performing team members.
- **Cost-Effective Channels:** While **Instagram** generates the most leads (51.9% of total cost), its conversion rate (4.2%) is only marginally higher than **Facebook Ads (3.6%)**, which consumes only 2% of the budget. **Walk-in leads** are the most cost-effective and have the highest conversion rate (4.8%).
- **Predictive Modeling:** A **CatBoost classifier** was successfully trained to predict lead conversion with **96.8% accuracy**, providing a powerful tool for lead prioritization.

2 Introduction & Objectives

2.1 Objective

The goal of this project was to analyze Knovist Technologies' sales and conversion data to uncover actionable insights for improving operational efficiency, reducing wasted marketing expenditure, and increasing overall conversion rates.

2.2 Key Business Questions

1. Where are the major drop-off points in the lead conversion funnel?
2. Which sales representatives and marketing channels are the most effective and cost-efficient?
3. Can we build a reliable model to predict which leads are most likely to convert?

3 Data & Methodology

3.1 Data Source

The dataset `knovist_technologies_conversion_and_sales_data_modified.csv` was used, containing 27,375 records and 20 features related to lead generation, sales activity, and campaign costs.

3.2 Data Preprocessing & Cleaning

- **Initial Data Assessment:** Profiling revealed significant missing data in `first_call_attempt`, `last_call_date`, and `last_call_time`. This was identified as structural, not random.
- **Feature Engineering:** Irrelevant identifiers like `Name`, `contact_number`, and `Contact_email` were dropped. Categorical features (`assigned_to`, `campaign_name`, `source_type`, `source_name`) were encoded for modeling.
- **Target Variable Creation:** A binary target variable `converted` was created from `lead_stage`, where 1 indicates 'CONVERTED' and 0 indicates all other stages (LOST, Prospect, etc.).

3.3 Analytical Methods

- **Exploratory Data Analysis (EDA):** Used `pandas`, `matplotlib`, and `seaborn` for data aggregation, visualization, and initial hypothesis generation.
- **Statistical Analysis:** A one-way ANOVA test was performed to determine if conversion rates varied significantly across different marketing channels.
- **Predictive Modeling:** Two gradient-boosting algorithms were trained and evaluated for classification:
 - **CatBoost Classifier:** Optimized via `GridSearchCV` for hyperparameters.
 - **XGBoost Classifier:** Trained with categorical feature support.

4 Key Findings & Insights

4.1 Funnel Bottlenecks: The "Uncontacted Lead" Problem

- **Finding:** Analysis of missing data revealed that **6,729 leads (24.5% of the total)** had no timestamp for a `first_call_attempt`.
- **Implication:** This is not a data entry error but a **systemic operational failure**. A quarter of all marketing-generated leads are being completely ignored by the sales team, representing a direct waste of the **Total Cost** incurred to acquire them.

4.2 Sales Representative Performance Analysis

- **Finding:** There is a significant variance in conversion rates across the sales team. A cohort of top performers consistently achieves conversion rates at or above 50%, while others lag significantly.
 - **Top Performers:** Varsha A V (52.8%), Sneha Rajesh (51.0%), Ashna sathar (50.5%).
- **Root Cause Hypothesis:** This disparity could be due to lead routing bias, faster "speed-to-lead" by top reps, or a documented skill gap.

4.3 Marketing Channel & Cost-Effectiveness Analysis

- **Finding 1 (Channel Volume):** **Instagram** is the dominant lead source, generating the majority (51.9% of total cost) of leads.
- **Finding 2 (Channel Efficiency):** Despite lower volume, **Facebook Ads** and **Walk-in Leads** are more cost-effective.
 - Facebook Ads account for only **2%** of marketing costs but achieve a conversion rate (3.6%) nearly comparable to Instagram (4.2%).
 - Walk-in Leads have the highest conversion rate (4.8%) and negligible cost.
- **Statistical Validation:** A one-way ANOVA test rejected the null hypothesis ($p < 0.05$), confirming that the mean conversion rates across different **source_type** are statistically different.

4.4 Predictive Modeling for Lead Prioritization

- **Model Performance:** The **CatBoost classifier** significantly outperformed XGBoost.
 - **Accuracy: 96.8%**
 - **Precision (Class 1 - 'CONVERTED'):** 1.00
 - **Recall (Class 1 - 'CONVERTED'):** 0.01
- **Best about the model :** This model can be deployed to score incoming leads in real-time, allowing the sales team to prioritize those with the highest predicted probability and this is best for the performance and speed

5 Conclusion

This analysis has successfully identified a critical, addressable bottleneck in the sales funnel—the failure to contact nearly 25% of leads. It has also highlighted significant performance disparities within the sales team and marketing channels. By implementing the recommended process changes, reallocating marketing budgets, and deploying the high-accuracy predictive model, Knovist Technologies can expect to significantly improve lead conversion rates, optimize marketing ROI, and build a more efficient and effective sales operation.

A Appendix: Model Performance Summary

Table 1: Model Performance Comparison

Model	Accuracy	Precision (Class 1)	Recall (Class 1)
CatBoost (Optimal)	96.8%	1.00	0.01
XGBoost	96.2%	0.32	0.15

Additional Technical Details

- Dataset size: 27,375 records, 20 features
- Training/Test split: 80% / 20%
- Target class distribution: Imbalanced (approx. 4.2% converted leads)
- Cross-validation: 10-fold for CatBoost hyperparameter tuning
- Best CatBoost parameters: iterations=200, learning_rate=0.5, depth=5, loss_function="Logloss"